Maithil Tandel

Data Scientist

+44 7767839535 | tandelmaithil@gmail.com | London, United Kingdom | GitHub | LinkedIn | Portfolio

Professional Summary

Data Scientist & Machine Learning Engineer specialising in **computer vision, image segmentation and deep learning**, with experience developing and optimising predictive models using Python, PyTorch and TensorFlow. Skilled in **object detection, image processing, feature extraction and semantic and instance segmentation**, with expertise in CNNs, OpenCV and transfer learning. Experienced in model evaluation, performance optimisation and translating complex visual data into actionable insights. Focused on building scalable, accurate computer vision solutions for real-world applications.

Skills

Functional Competencies: Computer Vision | Image Segmentation | Object Detection | Image Classification | Semantic & Instance Segmentation | Image Processing | Feature Extraction | Deep Learning | Model Training & Evaluation | Data Augmentation | Transfer Learning | Model Optimisation | Performance Evaluation | Statistical Analysis | Data Visualisation | Technical Documentation

Technical Tools: Python | OpenCV | PyTorch | TensorFlow | Keras | Scikit-learn | CNNs | U-Net | YOLO | Segment Anything Model (SAM) | NumPy | Pandas | Matplotlib | Git | Docker | SQL | Power BI

Work Experience

Data Scientist | **Checkatrade** | **London, UK (Hybrid)** *Feb 2026 – Present*

- Developed **machine learning and NLP models** using **Python, Scikit-learn and BERT** to detect fake reviews, classify customer feedback and identify deceptive review patterns.
- Built **Marketing Mix Models (MMM)** using regression, **adstock transformations and saturation curves** to quantify channel effectiveness, incremental sales and marketing ROI.
- Engineered **trade lead scoring models** using **XGBoost, Random Forest and Logistic Regression**, prioritising high-potential leads based on engagement, customer attributes and conversion probability.
- Implemented **multi-touch lead attribution models** using **Python and SQL** to evaluate marketing touchpoints, identify high-performing acquisition channels and analyse conversion journeys.
- Developed interactive **Power BI dashboards** to track model performance, marketing ROI, lead conversion rates and campaign effectiveness, supporting data-driven budget allocation and sales decisions.

Machine Learning Engineer | **EXCELR8 Motorsport** | **Nottingham, UK (Hybrid)** *May 2024 – Dec 2024*

- Built **forecasting models** using **Python** and **TensorFlow** on 10,000+ laps, achieving **95%+ accuracy** for race performance insights.
- Compressed telemetry data with **autoencoders**, cutting monthly storage by **60GB** and improving model pipeline throughput.
- Applied **MICE imputation** on **20%+ missing data**, enhancing predictive reliability and maintaining cross-race model stability.
- Clustered **300+ driver sequences** via **K-means**, improving team communication and consistency by **12%**.
- Built interactive **Power BI dashboards** to visualise predictions, supporting strategic setup for **25+ race events**.

Machine Learning Engineer | **Kennametal** | **Bengaluru, India (Onsite)** *Jan 2023 – Aug 2023*

- Built **CNN classifiers** for **4,000+ tools/day** using **TensorFlow**, cutting manual inspection by **3,000+ hours/year**.
- Designed real-time **anomaly detection** in **Python**, flagging **250+ defects/day** and improving inspection speed.
- Applied **predictive modelling** to **200+ metallurgy trials**, increasing R&D yield by **15%** through data-driven design.
- Centralised **2,000+ records** with **SQL** and visualised in **Power BI**, improving traceability and batch performance tracking.
- Automated deployment with **CI/CD pipelines** using **Git**, reducing model rollout from **14 days to 3**.

Deep Learning Engineer | **ResoluteAI.in** | **Bengaluru, India (Remote)** *Oct 2021 – Nov 2022*

- Developed **fraud detection models** on **500K+ transactions/day** using **Scikit-learn**, reducing risk exposure.
- Tuned models in **TensorFlow** to lower **false positives by 150+/day**, increasing fraud precision and trust.
- Built **BERT-based NLP models** on **50K+ tickets**, reducing resolution time by **28 minutes per case**.
- Scaled inference to **100K+/day** with **AWS Lambda** and **REST APIs**, maintaining sub-**300ms latency**.
- Created **Power BI dashboards** to monitor fraud patterns, enabling real-time decisions for risk analysts.

Computer Vision Research Engineer | **Smaaash Entertainment** | **Mumbai, India (Remote)** *Jun 2021 – Oct 2021*

- Built **pose estimation models** in **OpenCV** and **MediaPipe** with **92%+ accuracy** at **30 FPS**.
- Designed **DTW-based motion scoring** for **20+ moves/min**, enabling instant gameplay feedback and metrics.
- Reduced inference latency by **40%** via **model quantisation** and **pipeline optimisation for GPUs**.
- Documented **AR/VR architecture**, saving **28+ developer hours** in future multiplayer feature rollouts.
- Built **Power BI prototypes** for gameplay analytics, supporting product feedback and experience design.

Projects

Football Analytics | **Real-Time Object Detection & Tracking** | **Freelancer Project** *September 2025 - November 2025*

- Built a computer vision pipeline for **real-time detection and tracking** of players and the ball in soccer footage.
- Integrated **YOLOv8** for object & keypoint detection with **ByteTrack** for smooth, consistent multi-object tracking.
- Implemented **team classification** with annotated bounding boxes, posture keypoints, and confidence scores.
- Delivered an annotated demo video demonstrating applications in **sports analytics, coaching, automated replays, and broadcasting**.

Player Performance Profiling System (Tailored to Academy Analysis) | **Freelancer Project** *Feb 2025 - July 2025*

- Developing a data-driven tool for individual player analysis, demonstrating expertise in video and statistical insights for youth development.
- Analysed telemetry datasets using K-means clustering in Python to profile player styles, applying tactical knowledge to customise strategies and improve performance metrics by 17%.
- Created dashboards in Power BI for team syllabus review, integrating video clips to visualise transitions and reduce coaching feedback time by 25 minutes per session.

Personal Finance AI | Multi-agent system for real-time portfolio sentiment analysis *November 2025*

Tech Stack: Python, Agno, Gradio, Anthropic Claude, yfinance, DuckDuckGo Search, SQLite

- Built a multi-agent AI platform using **Claude API** and the **Agno framework** to analyze investment portfolios from live financial data.
- Designed agents like **News Reliability Analyst** and **Sentiment Synthesizer** to evaluate news credibility and compute sentiment scores.
- Developed an interactive **Gradio UI** for users to explore indicators like SMA, volatility, and sentiment shifts in real time.
- Enabled dynamic markdown report generation and SQLite-backed tracking of historical portfolio sentiment.

Grocery Product Recommender | Real-time semantic recommendation system for e-commerce search *October 2025*

Tech Stack: Python, NLP, Sentence-Transformers, FAISS, difflib, Git LFS, Hugging Face Spaces

- Engineered a semantic product search system using **sentence-transformers** and **FAISS** with sub-second response times.
- Increased search relevance by **25%** and user retention by **12%** through embedding-based ranking.
- Applied **difflib** for fuzzy matching and used **Git LFS** to manage 50MB+ datasets, improving valid coverage to **95%**.
- Deployed model on **Hugging Face Spaces**, achieving £500/year in infrastructure cost savings.

Restaurants Agent | Multi-agent platform for forecasting and rota automation *July 2025*

Tech Stack: Python, SQL, LLMs, Prophet, Automation, Predictive Analytics

- Built a forecasting tool using **LLMs** and **Prophet models** on 70K+ transactions from 3 restaurant sites.
- Automated rota creation for 12 employees using **role-fit algorithms**, reducing scheduling time by **80%**.
- Achieved **92% accuracy** in revenue prediction, enabling proactive staffing and supply planning.
- Delivered an SQL-backed dashboard and pipeline for easy integration with existing ops systems.

FashionSimilarity Engine | Vision-based product similarity for fashion discovery *June 2025*

Tech Stack: Python, PyTorch, CLIP, ViT, ConvNeXt, FAISS, Computer Vision, Jupyter

- Designed a visual similarity engine using **CLIP**, **ViT**, and **ConvNeXt** for 10K+ fashion items.
- Achieved **18% improvement** in visual match precision and delivered sub-second search with **FAISS indexing**.
- Developed modular **Jupyter notebooks** for model explainability, trend forecasting, and embedding exploration.
- Enabled fashion teams to cut styling overhead by **15%** through automated discovery and comparison.

Education

University of Nottingham | **MSc Machine Learning in Sciences** | **Nottingham, UK** **Sep 2023 – Sep 2024**

Relevant Modules: Neural Computation, Computer Vision, Segmentation, Predictive Modelling, Machine Learning, Big Data

Dissertation: *Optimizing Motorsport Performance: Predicting Ideal Lap Times and Extracting Feature Boundaries through Data-Driven Approaches Using Driving Style Patterns of the Racer*

Gujarat Technological University (GTU) | **B.E in Mechatronics, Robotics, and Automation** | **India** **Jul 2019 – Jul 2023**

Relevant Modules: Machine Learning, Statistics, Cloud Computing, Data Handling

Microsoft Learn Student Ambassador and Github Campus Expert

Dissertation: *Drone Photogrammetry – Built a drone-based 3D terrain mapping system integrating aerial image stitching, coordinate triangulation, and surface reconstruction for land survey applications.*

Publications

- **CORAL–Classification of Reefs and Analysis using Learning algorithms of image processing** | **Indian Journal of Geo-Marine Sciences (IJMS)** **December 2024**
- **ONNX for Underwater Plastic Detection** | **Open Source For You** **Aug 2023**
- **Multistep Data Driven Short Term Electrical Load Forecasting using LSTM** | **IEEE Xplore** **Feb 2022**

Certifications

- **TensorFlow Developer Certificate** | TensorFlow / DeepLearning.AI **Apr 2024**
- **Natural Language Processing Specialisation** | DeepLearning.AI / Coursera **Feb 2024**
- **Responsible AI** | Microsoft / DataCamp **Dec 2023**
- **Generative AI with LLMs** | DeepLearning.AI / Coursera **Nov 2023**
- **Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning** | Coursera **Jun 2021**
- **IBM Data Science Professional Certificate** | IBM / Coursera **Mar 2021**
- **Unsupervised Learning: Clustering** | upGrad **Jan 2021**
- **Data Analytics Using MS Excel** | The Sparks Foundation **Nov 2020**